HighGen—Comparative genomics resources for cannabis

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With **HighGen**, we adapted UCSC's chain/net and MULTIZ tools, tuned for cross-species vertebrates, in Galaxy across **217 *Cannabis sativa* assemblies** anchored on cs10—recalibrating repeat masking, empirical substitution scoring, chunked extension, and streaming column-count MULTIZ, paired with 43-genome Panagram *k*-mer bitmaps, a TOGA2 orthology pilot, and a 78-family phenotype ledger.

Galaxy

228-stanza (219-assembly) Hub
Attach depot.galaxyproject.org/highgen/hub.txt in the UCSC Genome Browser (Track Hubs → My Hubs).

A Four cross-species bottlenecks

In *Cannabis sativa*, interspersed repeats (63 to more than 70%, mostly LTR retrotransposons) surround dense tandem metabolic arrays (Grassa et al. 2021, Lynch et al. 2025, Pike et al. 2026). Projecting 216 query assemblies onto the linear cs10 (CBDRx) reference coordinate grid confronts four vertebrate-default obstacles:

1 Masking bias
Arrived soft-masking spans 0.00–80.86% across 218 assemblies, and KegAlign (Gulhan et al. 2025) seeds and extends ungapped only through uppercase bases, a seeding ascertainment bias

2 Cross-species matrix
LASTZ's default HOXD70 (Chiaromonte et al. 2002) was trained on human-mouse alignments below 70% identity, where 49.7% of pooled *Cannabis* pairs fall (Ts/Tv 0.872, chance ~0.74)

3 Superlinear scaling
Whole-chr LASTZ (Harris 2007) scales as $O(n^{1.9})$, while 217-way MULTIZ (Blanchette et al. 2004) overflows 32-bit scores and buffers MAFs in RAM

4 Macro-syntenic nets
Human-mouse netFilter -syn (Kent et al. 2003, maxFar = 200 kb) prunes rearranged tandem gene clusters

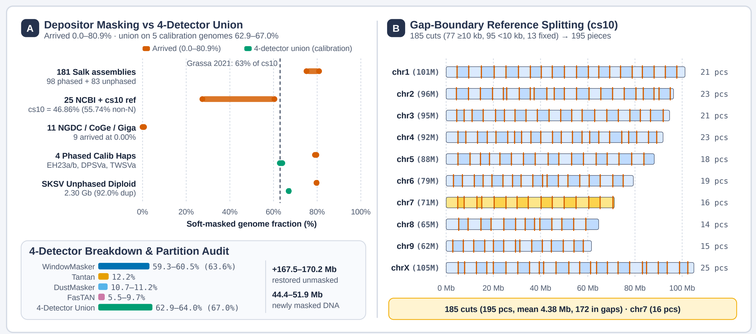


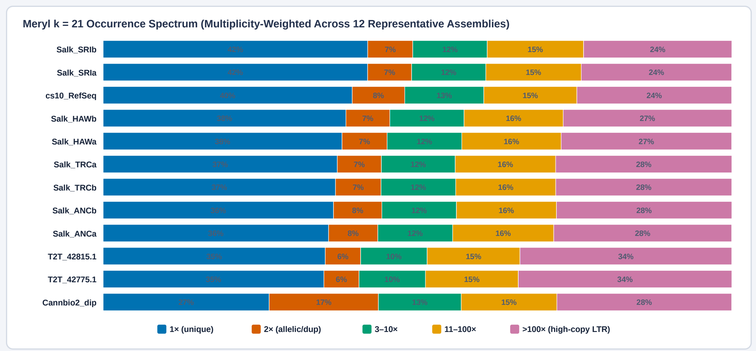
Figure 1 | Standardized masking and gap-boundary reference splitting. Arrived masks (0.0–80.9%, 218 assemblies) vs the 4-detector union on 5 calibration genomes (62.9–67.0%), grey line at the 63% of cs10 Grassa et al. (2021) masked (**A**), while 185 chromosomal cuts (77 at N-runs ≥ 10 kb, 95 at N-runs under 10 kb, and 13 fixed 5 Mb cuts, wfi_cs10_splits.bed) partition cs10 into 195 ~5 Mb pieces plus 26 heavy unplaced scaffolds, mitochondrion NC_029855.1, and small.00, with mafSplit cutting straddling blocks without base loss and maf_concat rejoining the pieces (**B**).

B Standardized 4-detector repeat masking

To replace incompatible depositor masks across our 221 curated assembly records (217 non-redundant in MULTIZ— cs10 plus 216 queries representing 164 prefix groups and 156 distinct plants, scored with rosales_odb12 BUSCO, Lynch et al. 2025, Pike et al. 2025, 2026), our brc-tools UDT wrappers strip arrived masks and unite four de novo detectors, WindowMasker (Morgulis et al. 2006a), DustMasker (Morgulis et al. 2006b), Tantan (Frith 2011), and FasTAN (Myers 2026):

fasta-uppercase → WindowMasker DustMasker Tantan FasTAN → bedtools merge

Assembly / Cohort	Arrived Mask	4-Detector Union	Partition Audit and Compute
181 Salk assemblies (98 phased + 83 unphased)	75.12–80.86%	**62.19–67.04%** (5-genome intake pilot, 4 Salk + 1 NGDC: 22.61 → 4.85 pt spread)	On calibration haplotypes, bases masked by depositors but not by our detectors make ~22% of each haplotype newly seedable (repeat status unmeasured)
25 NCBI GenBank + cs10	27.42–60.43% (cs10 46.86% total, 55.74% non-N)		Leaves 44.26% of cs10 non-N bases open to seeding
11 NGDC / CoGe / GigaDB	0.00–0.68% (9 kept at 0.00%)		Unmasked queries seed across all of their bases
4 Phased Calibration Haps (EH23a/b, DPSVa, TWSVa)	**79.00–79.61%**	**62.90–64.02%** (1.12-pt window)	**WM 59.30–60.53%, Dust 10.7–11.2%, Tantan 12.2%, FasTAN 5.5–9.7% · 3.64–3.99 CPU-h/hap**
Diploid SKSV (2.30 Gb, 92.0% BUSCO dup)	79.71%	**66.95%** (4.05-pt panel window, WM 63.60%)	432.1 Mb restored, 139.1 Mb newly masked · 10.34 CPU-h



12-genome Meryl *k* = 21 occurrence spectrum: Phased haplotypes (cs10, 8 Pike et al. 2026 ONT, 2 IMPD T2T) carry 35–42% 1× and 6–8% 2× multiplicity-weighted *k*-mers, whereas uncollapsed diploid Cannbio-2 (Braich et al. 2020) carries 27% 1× and 17% 2×.

Across phased haplotypes, the union restores 167.5–170.2 Mb of depositor-masked DNA while newly masking 44.4–51.9 Mb—a net unmasked gain of +115.9 to 123.2 Mb per haplotype with exact lowercase FASTA ↔ 0-based BED parity.

C Empirical intra-species substitution matrix

We adapted the Chiaromonte et al. (2002) log-odds protocol across two independent phased haplotype pairs (EH23a × EH23b and DPSVa × TWSVa, 95.3% rosales_odb12 BUSCO completeness, 1.8–2.0% duplication). Two of three steps restore her protocol: excising Salk GFF3 protein-coding CDS (–7.3% of chained segments) and single-chain colinearity (--max-chains 1). The departure is the inverted [85, 95)% training bracket. The colinearity filter enriched orthologous over non-orthologous matches **8.2-fold** on a Cb2 × cs10 slice (54.5% of the orthologous group's bases vs a 6.6% median across all 14 groups) and **18.0-fold** on EH23a × EH23b (mean per-group 11.1% vs 0.6% over the six largest orthologous and six largest non-orthologous groups).

331.05M
Pooled [85, 95)% pairs, symmetrized log-odds (Ts/Tv = 2.119, *H* = 1.4140 bits, either pair alone within 1 unit)

10/10 at 20 Mb
Wins vs HOXD70 on the unmasked first 20 Mb of all ten held-out chromosomes (chr8 rerun after OOM, 9/10 if counted a failure, $p = 0.0107$ vs $\alpha = 0.0083$), 9/10 loci, 67/70 windows (95.7%, 88.0–99.1% independent, 91.4–100% clustered)

5,033×
axtChain shuffled-null suppression (EH23a chr1 × *Morus*) + scaled 682/51 gaps (70.6% single coverage)

Normalized to max match 100 (+80 A/T, +100 C/G, –92 transitions, –191 A-C/G-T, –197 C-G, –198 A-T transversions), our matrix also won **10/10** at 20 Mb, **10/10 loci**, and **62/70 windows** vs unit ±100 across 40 false-match budgets. Its gain over the unchained [85, 95)% matrix was small, and stable only on the 20 Mb segments (10/10, margins 0.02–0.11%): on loci (9/10) and windows (55/70) the curves crossed, and the median stringent-end window margin (–0.16%) favoured the unchained matrix.

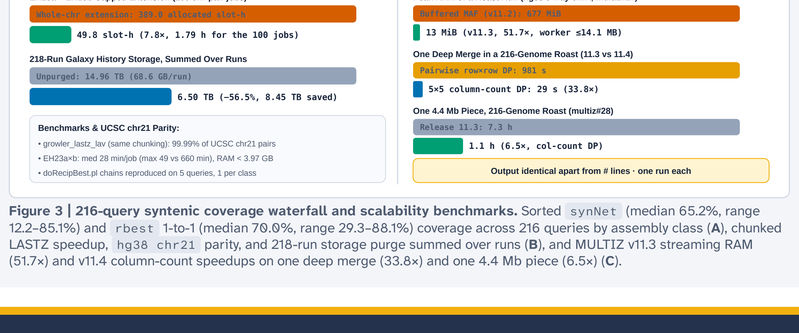


Figure 2 | Intra-species matrix calibration on the first 20 Mb of ten held-out chromosomes. EH23a × EH23b identity spectra before and after single-chain colinearity (**A**), wins of 40 false-match budgets vs HOXD70 on Cb2 × cs10 (whose chr1 chose the canonical matrix among candidates) on the first 20 Mb, the locus, and 70 eligible 1 Mb windows (**B**), and symmetric 4×4 log-odds weights with 682/51 gap scaling (**C**).

D Collection-native LASTZ and MULTIZ 11.3/11.4

- **Growler LASTZ (growler_ucsc_pair, LASTZ 1.04.52/1.04.60):** Couples KegAlign v0.3.4 GPU collection seeding (Gulhan et al. 2025) with indexed .2bit archives, compact LAV, and piped PSL chaining (Harris 2007, Kent et al. 2003). Batching anchor files into ≤50,000 -segment chunks cut growler_lastz gapped extension on EH23a × EH23b **7.8-fold** (389.0 → 49.8 slot-h, 1.79 h for the 100 jobs, 2.34 h invocation, median 28 min/job, under 3.97 GB RAM), 18% above the monolithic batched_lastz baseline (42.1 slot-h), with 5,166 seam-duplicated loci vs 2,135. The production wrapper growler_lastz_lav, which chunks identically, reproduced **99.99%** of UCSC's hg38 chr21 × panTro6 chr21 all.chain and rbest.chain pairs, and its doRecipBest steps reproduced doRecipBest.pl chains on one query of each of the five assembly classes.
- **Storage and 218-run production cohort:** Across all 218 runs (54,554 jobs, 10,325 job-hours, 113 on VGP and 105 on Laila), purging transient intermediate collections after workflow completion freed 8.45 TB, **56.5%** of the **14.96 TB** the histories held, summed over runs. Syntenic nets cover a median **65.2%** of cs10 non-N bases across 216 queries (range 12.2–85.1%, vs **70.0%** and 29.3–88.1% in rbest).
- **Streaming MULTIZ 11.3 and column-count 11.4:** Coordinate-ordered streaming cursors in v11.3 (multiz#24) cut the peak memory of a roast run on hg38 5-way chrX **51.7-fold** against v11.2 (677 → 13 MiB, multiz#27, 217-way worker RAM ≤14.1 MB). Its 64-bit path scores (multiz#19) changed 29 of 12,440 blocks of NC_044376.1.02 that the 32-bit build misaligned at exit 0, 24 of them scoring $\geq 2^{31}$ (Blanchette et al. 2004). Factoring mz_yama.c DP loops through column nucleotide counts (c_A, c_C, c_G, c_T, c_N) in v11.4 (multiz#28) accelerated one timed deep merge within a 216-genome roast **33.8-fold** (981 → 29 s against the 11.3 release) and cut a whole 4.4 Mb piece **6.5-fold** (7.3 → 1.1 h, one run each as reported upstream), with output identical apart from # header lines.

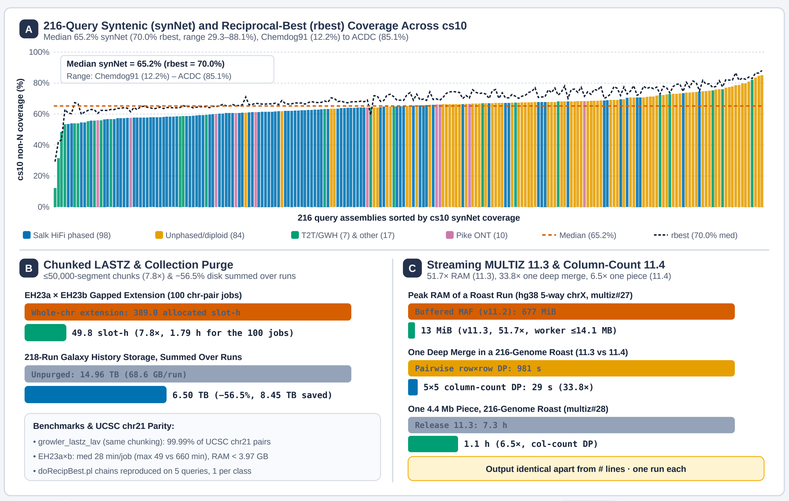


Figure 3 | 216-query syntenic coverage waterfall and scalability benchmarks. Sorted synNet (median 65.2%, range 12.2–85.1%) and rbest 1-to-1 (median 70.0%, range 29.3–88.1%) coverage across 216 queries by assembly class (**A**), chunked LASTZ speedup, hg38 chr21 parity, and 218-run storage purge summed over runs (**B**), and MULTIZ v11.3 streaming RAM (51.7×) and v11.4 column-count speedups on one deep merge (33.8×) and one 4.4 Mb piece (6.5×) (**C**).

E Tandem metabolic clusters and syntenic net limits

- **cs10 tandem census and permutation audit:** Adapting MCScanX tandem criteria (Wang et al. 2012) to stricter BLASTp thresholds ($E \leq 10^{-10}$, top 10 hits), **3,035** of cs10's 25,296 RefSeq protein-coding genes reside in 1,208 tandem arrays ($K = 0$) and **5,123 (20.3%)** in 1,796 clusters ($K \leq 10$), including **41 terpene synthases (TPS)** (31 in 9 clusters + 10 singletons) and **35 cannabinoid oxidocyclases (BBE-like)** (29 in 9 clusters + 6 singletons including CBDAS). Across 20,000 cluster-level permutations, nominal TPS low-identity enrichment (37.0% vs 11.0% controls, $p = 0.022$) does not survive family-wise correction ($p_{FWER} = 0.12$).
- **Functional CBDAS vs B-locus pseudogenes:** At cs10 functional CBDAS (LOC115697762, chr7:30.98 Mb, Grassa et al. 2021), our 217-way alignment separates **79 assemblies** (59 plants, including 56 Lynch full-length CBDAS) at ≥98.5% identity and **6 assemblies** below 93.0% (≥50% CDS) from **77 assemblies** (67 plants) at 93.0–98.5%. In 63 of the 77, Lynch et al. (2025) annotations place the alignment on the ~94% CBDAS-like B-locus pseudogene that Laverty et al. (2019) described near THCAS (62 of 63 in THC-dominant cultivars). The other 14, including two at 97.3 and 97.4%, carry no annotation.
- **netFilter -syn pruning vs paralog reuse:** UCSC applies its human-mouse distance and score cutoffs (maxFar = 200 kb, minTopScore = 300,000) only to contiguous assemblies and gives the rest reciprocal-best nets (Miller et al. 2007). Applied beyond that rule to all 216, including at least 65 that fail a scaffold-N50 contiguity test, it pruned **4.7% of TPS** (418/8,856) and **10.7% of BBE-like** (807/7,560)—dropping 54 CBDAS assemblies (under 50% CDS) in synNet, including Finola's intact 99.4% transposed copy (CM011610.1:21.84 Mb, 26.8 Mb from its enclosing gap). Of the 1,225 alignments the filter's removal recovers, **46.4%** (569) overlap another family gene's query interval, against 10–21% of those it keeps. Resolving rearranged clusters thus needs rbest nets or recalibrated thresholds with duplication-aware orthology such as TOAST and BOAST (Hou et al. 2006, 2009).

UCSC Hub: cs10 chr7 (25.02–31.03 Mb) CBCAS and CBDAS cassettes



UCSC Hub: multiz217way at CBDAS (intact, ~94% B-locus, and pruned rows)



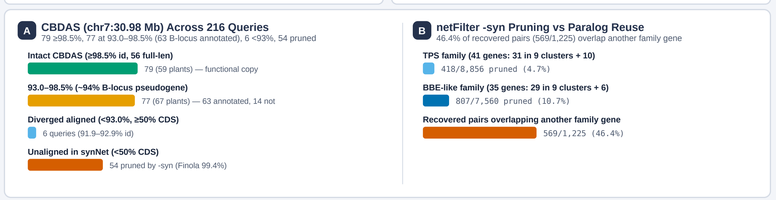


Figure 4 | Live HighGen UCSC Track Hub on chr7 and netFilter -syn trade-off. Browser views of the 6 Mb cannabinoid synthase super-cluster (left) and multiz217way at CBDAS (right), paired with 216-query CBDAS resolution (**A**) and netFilter -syn pruning beside the 46.4% of recovered pairs that overlap another family gene (**B**).

F Panagram bitmaps, TOGA2 pilot, and phenotype ledger

- **Panagram v1.1.0 pan-*k*-mer indexing (panagram_index / panagram_view, Jenike et al. 2026, Kokot et al. 2017, Aylward et al. 2023):** Tier P1 (17 genomes: cs10 + 16 Salk haplotypes from 9 plants, 13.3 Gb) selected $k = 21$ over $k = 31$ (higher linear 1 Mb window Pearson r with rbest identity in **16/16 Salk haplotypes**, mean $r = 0.7150$ vs 0.6901, 16/16 higher at $k = 21$). Tier P2 across **43 genomes** (32.64 Gb, 5.05 h on 16 Laila slots, 1.14 GB index) passed $r \geq 0.50$ in **41 of 42 chain-scored query genomes** ($r = 0.5848$–0.8118, including Pink pepper at $r = 0.7376$), passed non-PAR chrY separation (3/3 males 0.976–0.987 vs 3/3 V3 females ≤ 0.077 across 205,619 100-bp-subsampled AH3Ma -absent positions), flagged 2 higher-QV Guppy 5 SUP trio-binned ONT haplotypes on G3 count tolerance (Punto Rojo –10.4%, Cherry Pie –10.8%, Pike et al. 2025), and saw only one Sri Lankan ONT haplotype (ASM5191173v1, $r = 0.4763$, Merqury QV 24.01/24.11, Pike et al. 2026) fall below 0.50, for a cause still open.
- **TOGA2 v2.0.9 orthology pilot (Kirilenko et al. 2023, Malovichko et al. 2026):** On the cs10 reference annotation, we stratified all **25,261** nuclear protein-coding genes (33,639 transcripts, excluding mitochondrion NC_029855.1) into **24,503 standard** and **758 reference-anomaly** genes alongside 221 inventoried all.chain datasets (1.45 TB). On its zero-divergence identity-chain control, TOGA2 called **25 of 24,503 lost/uncertain (0.10%**: 1 L + 24 UL against a ≤ 0.5% bar). On EH23a, the production, loose, and RepeatFiller -filled-and-cleaned chains called **2,077 (8.48%)**, **1,778 (7.26%)**, and **1,660 (6.77%)** standard genes L/UL. The filled and production arms disagree on the class of 713 of the 2,161 genes either calls L, UL, or M (33.0%). On false presence, the second of the plan's two ranking criteria, production ranks first (4,306 standard genes with more than one query gene, against 5,058 loose and 5,245 filled). The first criterion is not yet implemented, so the arms are not ranked, no arm is chosen, and no gene loss is reported.
- **Provenance-verified phenotype ledger:** Joining 30 publications (43 audited extractions, including 988 Lynch et al. 2025 Figshare rows) and 10 NCBI BioSamples across 215 assemblies yielded **1,341 trait records** across **154 plants in 78 pedigree families** (69 plants in the largest Salk family, 71 singletons with no pedigree link we parsed). A first between-family test of a pre-specified candidate needs at least 30 unrelated plants (one per family), a conventional floor at which the smallest attainable p is 7×10^{-6}. Under strict measured / reported filtering, none of 14 traits yet meets it (Varin comes closest at 30 families, but 37 of 38 non-WT plants derive from HO40).

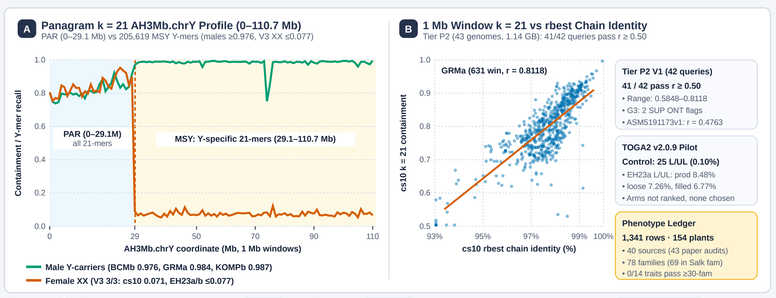


Figure 5 | Alignment-free Panagram chrY profile and rbest correlation. Chromosome-wide AH3Mb.chrY (0–110.7 Mb) $k = 21$ profile showing PAR all-21-mer containment (0–29.08 Mb) and MSY male-specific 21-mer recall across 205,619 subsampled positions (**A**), and all 631 eligible 1 Mb windows on GRMa ($r = 0.8118$) with Tier P2, TOGA2, and phenotype scorecards (**B**).

Public tools, workflows, and 228-stanza (219-assembly) browser hub

4-detector masking → brc-tools · GPU seeding + chunked LASTZ → growler_lastz_lav · 217-way streaming TBA → MULTIZ 11.3/11.4 · Pan-*k*-mer + orthology → Panagram and TOGA2

Availability: HighGen 228-stanza UCSC Track Hub (219 unique assemblies across 157 plants + 9 alias stanzas: multiz217way.bigMaf.bb, reading frames, pairwise chains/nets, rbest chains, gene models, and repeat masks): https://depot.galaxyproject.org/highgen/hub.txt (each assembly's hub page names its source and original location and gives a citation and a licence line, four pages lack a data holder, and 9 CoGe/NGDC redistribution grants are pending per assembly_licences.md). Analysis repository, scoring matrix, and scripts (licence pending): https://github.com/richard-burhans/cannabis-genome. Galaxy tool wrappers (panagram_index, panagram_view, growler_lastz, gt-multiz-roast) and workflows (growler_ucsc_pair, wfi_split, wfi_roast, wfi_track): https://github.com/richard-burhans/galaxytools (MIT) and https://github.com/nekrut/brc-tools (masking workflow, licence pending) · MULTIZ 11.3/11.4: https://github.com/multiz/multiz · LASTZ 1.04.52/1.04.60: https://github.com/lastz/lastz · KegAlign 0.3.4: https://github.com/galaxyproject/KegAlign · Panagram 1.1.0: https://github.com/kjenike/panagram · TOGA2 2.0.9: https://github.com/hillerlab/TOGA2.

HighGen Hub · Galaxy Tools · Repository